TOPIC 2 · MICROSOFT COPILOT FOR EXCEL
LAB 5 THEME · CUSTOM PARETO CHART SKILL

Lab 5 — Create a Custom Pareto Chart Skill and Apply It

Maps to	LO2: plan a controlled transition using Personalisation, .Rules and reusable Skills.
Duration	30 minutes
Tools	Excel Copilot · Microsoft 365 Copilot web · OneDrive
Mock data	data/Lumina_HR_Cases.xlsx
Lab files	data/templates/hr-case-pareto/SKILL.md (Reviewed Pareto Skill fallback)
Course	Enhance Work Productivity with Microsoft 365 Copilot (TGS-2024043856) · v9.4
Prompts	https://tertiarycourses.github.io/TGS-2024043856-Enhance-Work-Productivity-with-Microsoft-365-Copilot/prompts/#lab-5

Goal

Author hr-case-pareto/SKILL.md with Copilot, install it in Excel and apply it to the same overdue HR-case queue. Build a descending count chart plus cumulative-percentage line, then check a controlled rerun and a second grouping. Contribution analysis does not establish root cause or force an 80/20 split.

What You Will Do

Author hr-case-pareto/SKILL.md with Copilot, install it in Excel and apply it to the same overdue HR-case queue. Build a descending count chart plus cumulative-percentage line, then check a controlled rerun and a second grouping. Contribution analysis does not establish root cause or force an 80/20 split.

What You Will Build

hr-case-pareto/SKILL.md, Lumina_Pareto.xlsx with Pareto_Request_Type and Pareto_Department, and a verified rerun record.

Prerequisites

- Microsoft 365 account with Copilot available in Word, Excel, PowerPoint and Outlook. Availability depends on the assigned licence and tenant settings; confirm access with the trainer before class.
- Desktop Excel on Windows or Mac is recommended; save working copies to your assigned OneDrive course folder with AutoSave on. Do not overwrite the supplied input.
- Download this whole lab folder. All files are synthetic; Lumina-HR-v9 is your own learner output folder.
- Lab 4 visible .Rules and clean source; assigned Excel Skill feature checked.

Data note

Lumina Living Pte Ltd and all of its employees, HR cases and HR policies are fictional. Use the verified 120-case baseline and fixed reporting date 30 September 2026. Current case owners are evidence; proposals are not approved. The HR agent answers from approved knowledge only and never approves requests, sends messages or reveals another employee's data.

Steps

Step 1. Open your rules-enabled workbook from Lab 4 or this lab's clean rules-enabled data/Lumina_HR_Cases.xlsx. Save as Lumina_Pareto.xlsx. Read data/Pareto_Expected_Controls.json and the supplied Skill template before drafting your own file. Work only with synthetic data.

PROMPT / CHECK — copy and adapt

```
EXCEL - CHAT ONLY:
For tblHRCases at Control!B2, All departments, count overdue backlog by Request_Type: Status
neither Closed nor Cancelled and Due_Date strictly before the reporting date. Show category
counts and total before charting. Preserve all records.
```

Step 2. In Microsoft 365 Copilot on the web at https://copilot.cloud.microsoft, provide this self-contained specification. Copy the complete response into a plain-text editor and save UTF-8 SKILL.md inside a matching hr-case-pareto folder. Compare it to data/templates/hr-case-pareto/SKILL.md. If short of time, use the reviewed supplied file and record that choice.

PROMPT / CHECK — copy and adapt

```
MICROSOFT 365 COPILOT WEB:
Write a complete SKILL.md named hr-case-pareto for a recurring HR service report. Require YAML
name and description, actual source table, reporting date, scope, count metric and grouping
column. Source tblHRCases has Case_ID, Request_Type, Department, Status and Due_Date. Overdue
backlog excludes Closed/Cancelled and has Due_Date before the supplied date. Read visible
workbook .Rules; preserve source IDs/statuses/owners and unrelated sheets. Count by grouping,
sort count descending then label alphabetically for ties, and create formula-backed count,
cumulative count, share and cumulative percentage. Build an editable combo chart: count
columns and cumulative line on a 0%-100% right axis. Output only Pareto_<grouping column>; ask
before replacing the matching output, rerun without duplicates, and leave zero-total
percentages N/A. Verify counts, stable ties and final 100%; do not infer causes or force
80/20. Return the entire file.
```

Step 3. In Excel use Copilot Settings (...) > Manage plugins & skills > Custom skills > Create OneDrive folder if needed > Open skills folder. Copy hr-case-pareto/SKILL.md into that app-created location, Refresh and enable it. Verify the name using Choose plugins & skills or @mention. If unavailable, paste the complete reviewed Skill before each request, remove @hr-case-pareto and record Manual practice - native Skill unavailable. Log web authoring manually.

Step 4. Request a plan before building. Confirm the eligible total 50 and the five ordered category counts. After review, apply the Skill. Inspect Pareto_Request_Type, formulas, axes and source preservation; copying a static image does not meet the editable-chart goal.

PROMPT / CHECK — copy and adapt

```
EXCEL - PLAN:
@hr-case-pareto Plan an overdue-backlog Pareto Chart by Request_Type from tblHRCases at
2026-09-30, All departments. Follow visible .Rules. Explain summary formulas, descending
counts, alphabetical ties, cumulative percentages and the named output to create. Do not edit
yet.

EXCEL - ALLOW EDITING:
@hr-case-pareto Apply the reviewed Request_Type Pareto plan. Create Pareto_Request_Type with a
formula-backed summary and editable count-column/cumulative-percent-line combo chart. Keep
source data and unrelated sheets unchanged.
```

Step 5. Check counts against the supplied controls. Verify cumulative percentages 36%,60%,76%,92%,100% and two equal8-count categories in alphabetical order. The first four categories reach at least80%; do not call this proof that20% of categories cause80% of failures. If combo chart UI is absent, select the summary and use Insert > Combo > Custom Combination: Count clustered column, Cumulative_Percent line on secondary axis, right bounds 0 and 1; label both axes. Record a missing feature honestly.

PROMPT / CHECK — copy and adapt

```
EXCEL - CHAT ONLY:
Verify Pareto_Request_Type against eligible source rows. Check total 50, descending counts, alphabetical tie order, cumulative percentages 36/60/76/92/100%, final 100% and labelled dual axes. State which category set first reaches 80%, and why this is contribution rather than a causal diagnosis.
```

Step 6. Save a sandbox copy, rerun the same grouping after approving replacement of only Pareto_Request_Type, and compare source IDs/statuses/owners and sheet/chart counts. Then apply the Skill to Department to produce a distinct Pareto_Department output. Keep both reviewed summaries for later decision evidence.

PROMPT / CHECK — copy and adapt

```
EXCEL - PLAN:
@hr-case-pareto Refresh the existing Pareto_Request_Type for the same source, date and All scope. Identify exactly what will be replaced and ask before replacement. Preserve every unrelated sheet and do not duplicate charts.

EXCEL - ALLOW EDITING:
@hr-case-pareto After the approved category rerun, create a separate overdue-backlog Pareto_Department for the same source/date/scope. Preserve Pareto_Request_Type and reconcile the new summary to 50 overdue cases.
```

Sample Prompts

Use the app labelled for each prompt (or adapt it). Name the table and columns, state the output, and always verify the result. Use the linked prompts web page for one-click Copy buttons.

Copy-ready prompts: https://tertiarycourses.github.io/TGS-2024043856-Enhance-Work-Productivity-with-Microsoft-365-Copilot/prompts/#lab-5

#	Purpose	Prompt
1	Run Skill	Apply @hr-case-pareto to Lumina_HR_Cases.xlsx for overdue work at 30 September 2026; verify category counts 18, 12, 8, 8, 4 and cumulative 100%.
2	Zero case	Run the zero-overdue test and return N/A rather than dividing by zero.

Test It

Check	Expected result
Eligible population	Overdue backlog50; no Closed or Cancelled records included.
Category counts	Payroll Query=18; Benefits Claim=12; Leave Request=8; Training Request=8; Onboarding=4
Cumulative percentages	36%,60%,76%,92%,100%; Leave Request precedes Training Request in the8-count tie.
Chart and rerun	Editable columns plus cumulative line; right axis0%-100%; matching output replaced only after review, without duplicates.
Source and interpretation	120 original records preserved; Department grouping also totals50; no invented root cause.

Checkpoint

If you fell behind, use data/checkpoints/Lumina_Verified_Checkpoint.xlsx and data/checkpoints/Lumina_Verified_Snapshot.docx. Record checkpoint used in Prompt_Log; then perform this lab, rather than claiming the checkpoint as your output.

Troubleshooting

If this happens	Fix
Copilot cannot find the source	Use the app file picker to select the actual OneDrive file. A typed local path or a different app chat is not shared context. If needed copy only the verified snapshot into the current app.
A total disagrees with Control	Clear dashboard filters to All, check table extent and duplicate IDs, recalculate, then compare definitions and reporting date. Reconcile before continuing.
The named feature is missing	Confirm the correct assigned account, English display language and app version. Use the stated manual practice route; record feature unavailable. This is not evidence that a native skill ran.

Challenge

Repeat the task for one different department and explain how the filter scope changes the result. Keep the original verified output.

Reflection

Which evidence proves the output is correct, and which decision still needs a person?